

Physics-Informed Machine Learning for Predicting Supporting-Boundary Circularity at Any Tool Position in Micro-EDM of Metallic Lattices

Lattice Circularity Analyzer Project Team

Research Paper — Ideation, Data Expansion, Training/Validation, Position-Aware Prediction

Abstract

We present a physics-informed machine-learning method that predicts the circularity ratio of the supporting material boundary after micro-EDM of a metallic lattice, for **any tool landing position** (x, y) and any EDM recipe (peak current I , pulse-on time T , duty factor D). Only sixteen laboratory runs with SEM labels were available. From these we generated 1,100 synthetic EDM points via a Gaussian Process posterior, and expanded each labeled run across a spatial landing grid by modulating SEM circularity with a geometry-risk penalty (≈ 400 position-aware training rows). Dual Gradient Boosting regressors learn circularity (1–5) and supporting integrity; predictions are blended with EDM physics heuristics. Circularity ratio = score/5. SEM evidence shows only Run 4 (4 A, 150 μ s, 80%) yields an intact circular supporting ring — the objective the models maximize.

Keywords: micro-EDM; lattice structures; circularity ratio; Gaussian process; gradient boosting; physics-informed ML; data augmentation; SEM validation

1. Problem Definition

A metallic lattice has square unit cells of side 500 μ m. Pore and node diameters equal $x = 500\sqrt{2} / 3 \approx 235.6$ μ m. The EDM tool tip is 900 μ m (tool/pore = 3.82), so the electrode always overlaps pores, nodes, and supporting struts together. Success means a nearly circular open region with a **continuous supporting ring**; nodes may be destroyed. Hole-deviation numbers alone are insufficient: among sixteen SEM-validated trials, only Run 4 produced the desired ring.

The question answered here: given (I, T, D, x, y) , predict circularity ratio and whether the supporting boundary survives — even at positions never machined in the lab.

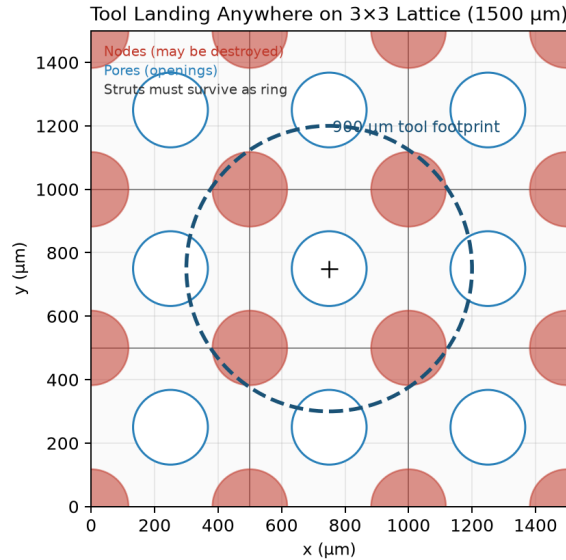


Fig. 1. Tool footprint (900 μ m) on a 1500 $\times$ 1500 μ m working area. Landing position changes strut/node overlap and circularity risk.

2. Ideation — How Circularity Can Be Predicted Anywhere

Circularity is not a property of EDM parameters alone. It is the **joint effect of process intensity and local lattice geometry under the tool**. Identical (I, T, D) can succeed at a pore center and fail near a strut. That coupling is the working idea of the solution.

Idea 1 — Train on SEM boundary truth. Do not minimize Hole_Dev_Top/Bottom. Maximize supporting-boundary circularity from SEM. Run 5 looks best by deviation but destroys the ring; Run 4 is the only SEM success.

Idea 2 — Geometry is a first-class feature. Before ML, a lattice engine computes how the tool intersects nodes, pores, and struts at (x, y) .

Idea 3 — Expand sparse lab truth with physics structure. Densify EDM space with GP sampling, and replay each SEM label across a landing grid with geometry penalties.

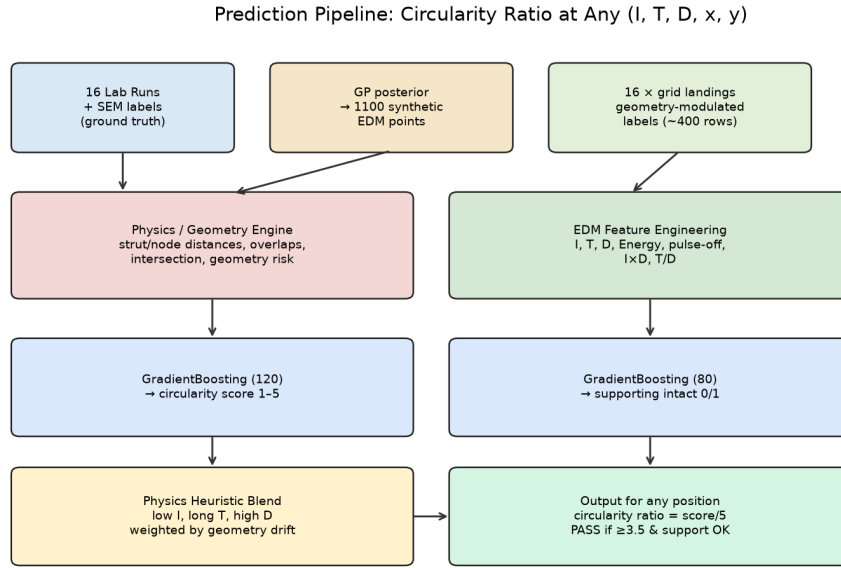


Fig. 2. Prediction pipeline: SEM truth $\rightarrow$ dual augmentation $\rightarrow$ physics + EDM features $\rightarrow$ Gradient Boosting + heuristic blend $\rightarrow$ circularity ratio at any position.

3. Experimental Evidence — Graphs from the 16 Runs

All charts below are computed from the laboratory table and SEM labels. They show why deviation-only ranking misleads and why the ML target must be SEM circularity.

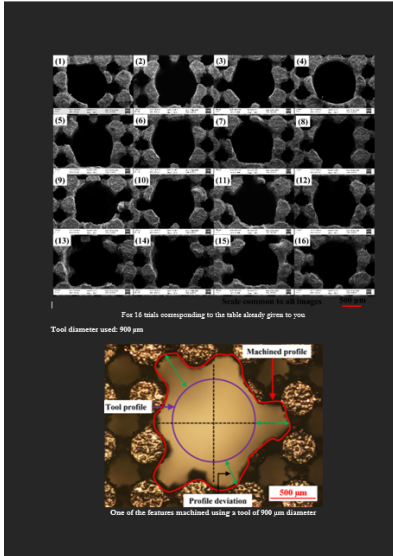


Fig. 3. SEM montage of all 16 experimental outcomes (visual ground truth).

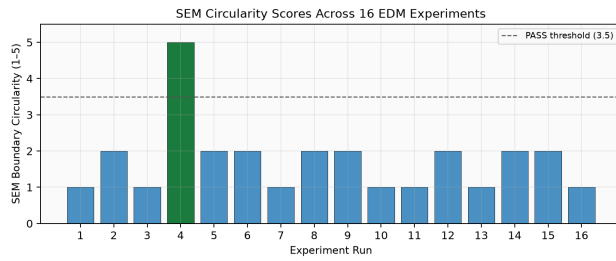


Fig. 4a. SEM circularity by run (green = PASS).

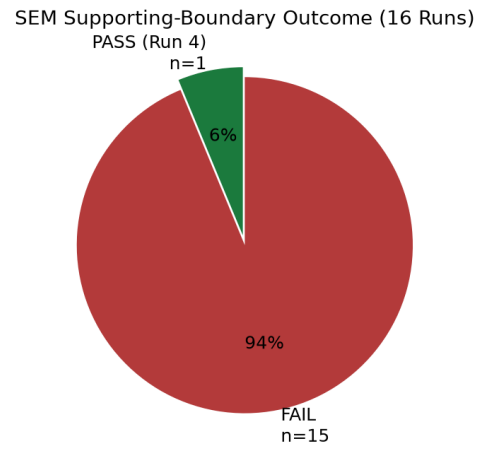


Fig. 4b. 1 PASS / 15 FAIL supporting-ring outcomes.

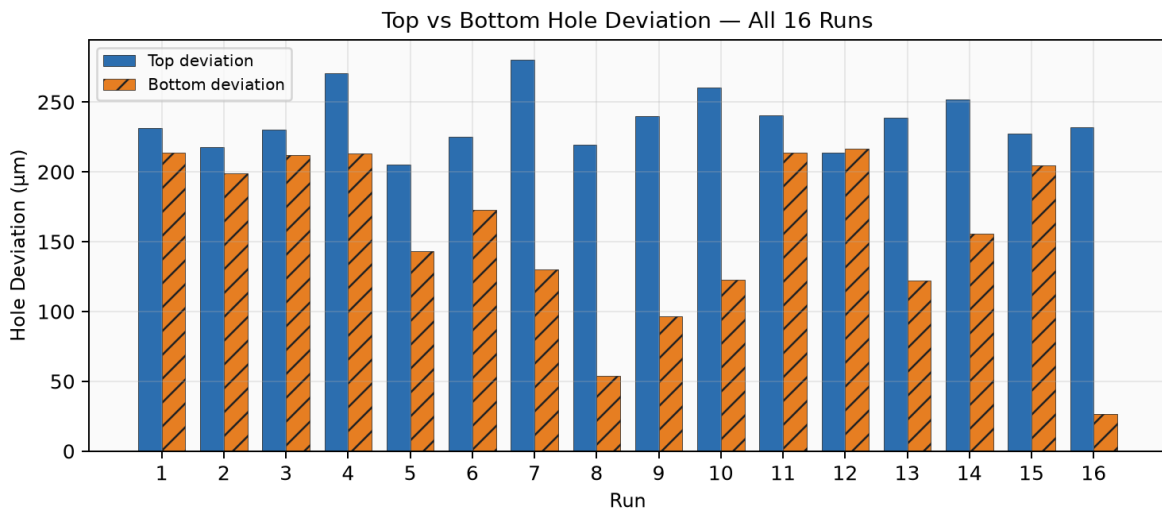


Fig. 5. Top vs bottom hole deviation for all 16 runs. Large top–bottom gaps indicate taper (asymmetric plasma erosion).

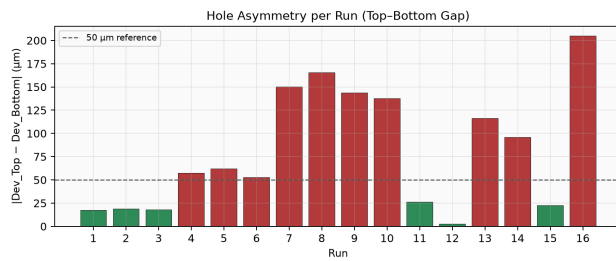


Fig. 6a. Absolute asymmetry $|Dev_Top - Dev_Bot|$.

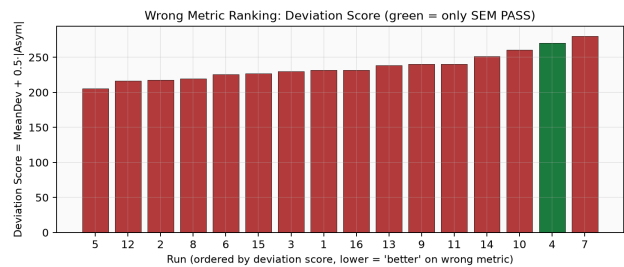


Fig. 6b. Deviation-score ranking (wrong metric): Run 4 near worst, yet only SEM PASS.

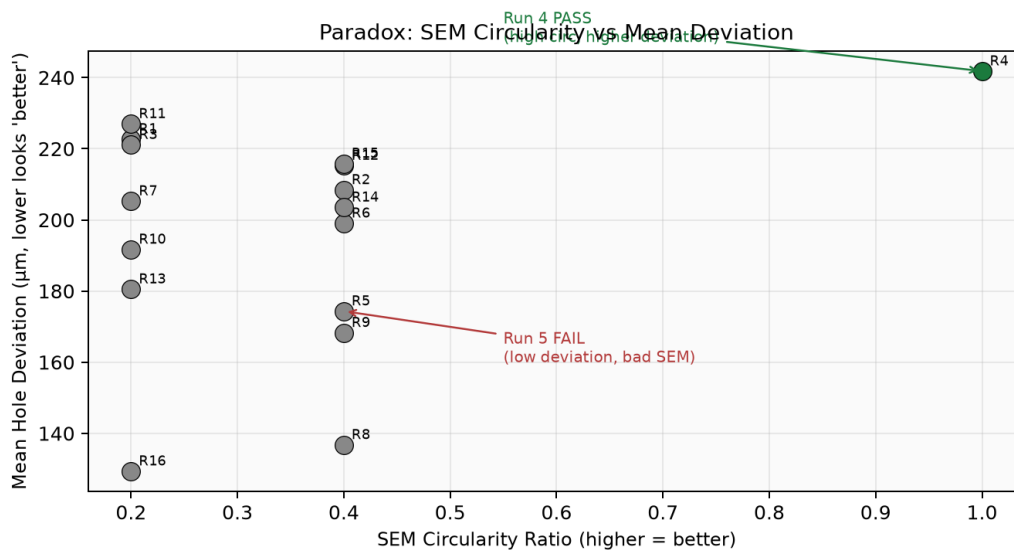


Fig. 7. Paradox scatter: SEM circularity ratio vs mean deviation. The SEM PASS (Run 4) does not sit in the low-deviation corner.

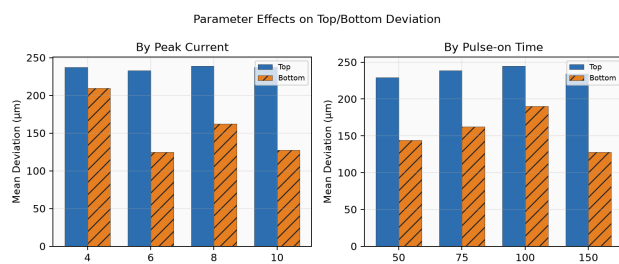


Fig. 8a. Mean top/bottom deviation by current and pulse-on.

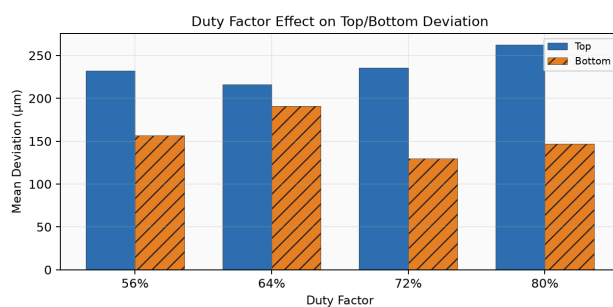


Fig. 8b. Duty-factor effect on top/bottom deviation.

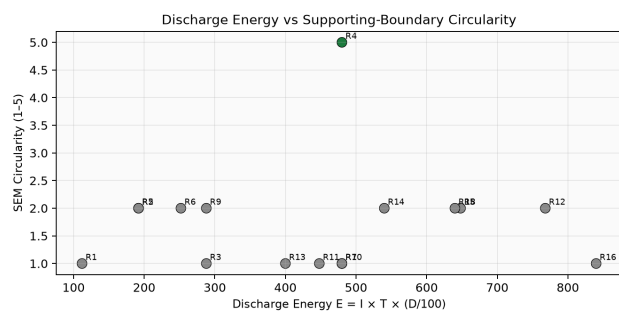


Fig. 9a. Discharge energy vs SEM circularity.

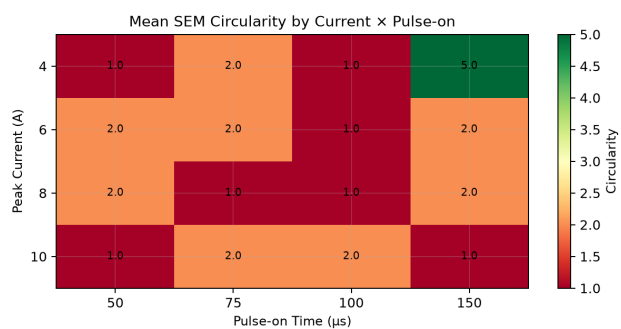


Fig. 9b. Mean circularity heatmap: Current x Pulse-on.

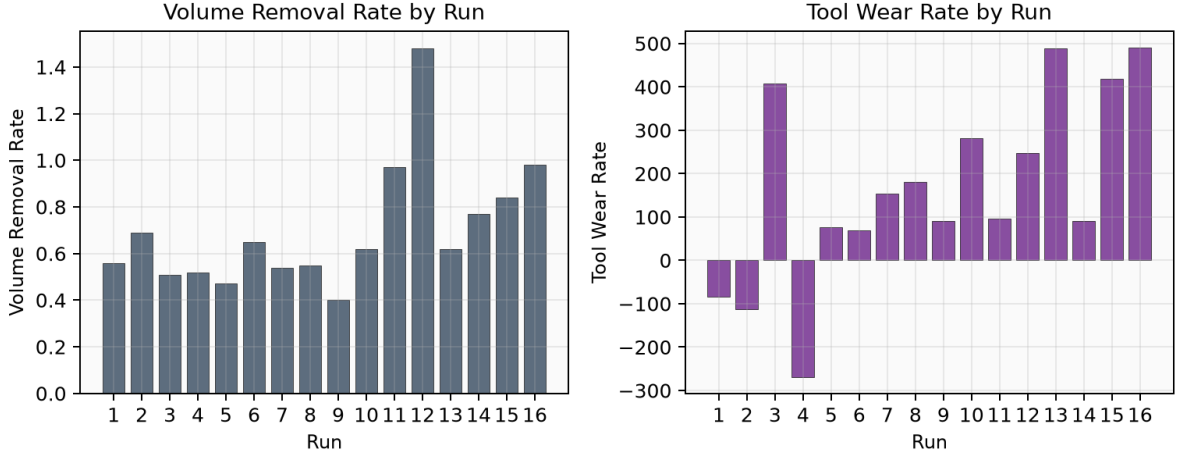


Fig. 10. Process responses: volume removal rate and tool wear rate across the 16 runs (context for intensity, not the success metric).

4. Physics Used in the Predictor

EDM descriptors: discharge energy $E = I \cdot T \cdot (D/100)$, pulse-off $\approx T \cdot (100-D)/D$, $I \times D$, T/D . Heuristics reward low current / long pulse / high duty and penalize $I \geq 8$ A.

At landing (x, y) the geometry engine computes min distances to strut/node, nodes/pores inside the tool, strut intersection length, pore/node overlap fractions, and

$$\text{geometry_risk} = 0.5 \cdot \text{strut_risk} + 0.3 \cdot \text{intersect_risk} + 0.2 \cdot \text{ratio_factor} \in [0,1].$$

5. From 16 Experiments to a Trainable Dataset

Augmentation A — GP synthetic EDM points (1,100). A Gaussian Process fit on the sixteen real $(I, T, D) \rightarrow$ response mappings was sampled in-bounds to produce `synthetic_1100_points.csv` (source `GP_posterior`). These densify process space; they are not new lab experiments.

Augmentation B — Spatial label replay (~400 rows). Each SEM label is applied on a landing grid (≈ 25 positions at $150 \mu\text{m}$ step inside valid tool-center bounds):

$$c_{\text{train}} = \text{clip}(c_{\text{SEM}} - 2.5 \cdot \text{geometry_risk} + \text{edm_bonus}, 1, 5).$$

Gentle recipes get a small bonus; $I \geq 8$ gets a penalty. This teaches that the same recipe degrades when the tool sits on dense strut intersections.

How 16 Experiments Become Thousands of Training Rows

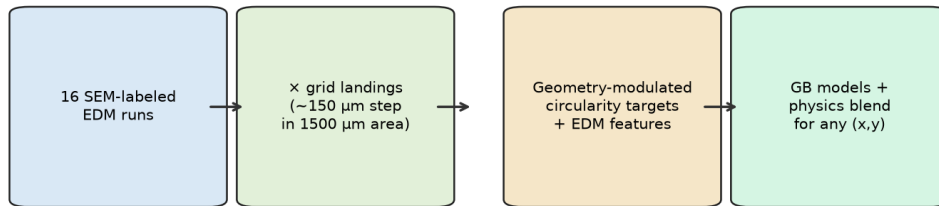


Fig. 11. Data-expansion path from 16 SEM-labeled runs to position-aware training rows.

6. Machine Learning in Detail

This section explains exactly what is learned, from which labels, with which algorithms, how models are validated on only sixteen SEM trials, and how a prediction is formed for an unseen landing position.

6.1 Learning targets (what the models try to predict)

Two supervised targets are used. **Target A — boundary circularity score** $c \in \{1,2,3,4,5\}$ from SEM judgment of the supporting ring (Run 4 = 5; most failures = 1–2). At inference we also report **circularity ratio** = $c/5$ (so $3.5/5 \rightarrow 0.70$). **Target B — supporting integrity** $s \in \{0,1\}$: whether the black ring is continuous. Hole deviation is *not* the training target for the position-aware system, because minimizing it recovers Run 5 and destroys struts.

6.2 Feature vector (20 dimensions)

Every training row and every query builds the same 20-D vector $\mathbf{X} = [\mathbf{X}_{\text{EDM}} \mid \mathbf{X}_{\text{geom}}]$.

EDM block (7 features): raw settings I, T, D ; discharge energy $E = I \cdot T \cdot (D/100)$; pulse-off proxy $T \cdot (100-D)/D$; interaction terms $I \cdot D$ and T/D . These terms let trees split on intensity and on “gentle vs punchy” delivery, not only on raw ampere or microsecond values.

Geometry block (13 features): normalized landing $x/W, y/W$ ($W = 1500 \mu\text{m}$); min distance to strut; min distance to node center; count of nodes inside the tool; count of pore centers inside the tool; strut intersection length inside the footprint; pore overlap fraction; node overlap fraction; scalar geometry risk; tool/pore ratio; working area; tool diameter. This block is why the model can answer “same recipe, different (x, y) ”.

#	Feature	Type	Role in learning
1–3	I, T, D	EDM raw	Primary process controls
4	Energy E	EDM derived	Total spark intensity
5	Pulse-off proxy	EDM derived	Flush / recovery time
6–7	$I \cdot D, T/D$	EDM interaction	Duty-coupled effects
8–9	$x/W, y/W$	Geometry	Where the tool lands
10–11	Dist. strut/node	Geometry	Clearance to ligaments
12–13	Nodes/pores in tool	Geometry	How much cell is engulfed
14–16	Strut len, overlaps	Geometry	Damage exposure
17	Geometry risk	Geometry	Compressed risk index
18–20	Ratio, W , tool $\emptyset$	Config	Scale / overfill context

Table 1. Full 20-D feature vector used by Gradient Boosting.

6.3 Building supervised labels from only 16 SEM runs

Lab SEM gives one circularity and one integrity flag per EDM recipe — not per landing. To supervise position dependence we **replay** each recipe on a landing grid (step $150 \mu\text{m}$ inside valid tool-center bounds $\rightarrow$ 25 landings $\rightarrow$ $16 \times 25 = 400$ rows):

$$c_{\text{train}}(\text{run}, x, y) = \text{clip}(c_{\text{SEM}}(\text{run}) - 2.5 \cdot \text{geometry_risk}(x, y) + \text{edm_bonus}(I, T, D), 1, 5)$$

where $\text{edm_bonus} = +0.5$ if $(I \leq 5, T \geq 130, D \geq 75)$, -1.0 if $I \geq 8$, else 0. Supporting label $s_{\text{train}} = 1$ only if the SEM flag is intact *and* $\text{geometry_risk} < 0.6$ *and* $I < 8$; for near-Run 4 recipes $(I \leq 4.5, T \geq 140, D \geq 78)$ we allow $s_{\text{train}} = \max(s, 1 - \text{geometry_risk})$. This is physics-informed label propagation: the SEM score is the anchor; geometry risk lowers the target when the tool sits on struts.

Separately, a Gaussian Process posterior over the 16 real (I, T, D) points produces 1,100 synthetic EDM rows used to densify process-parameter space for exploratory models. Those synthetics are tagged **GP_posterior** and are never treated as new SEM ground truth.

6.4 Algorithms and hyperparameters

(A) Position-aware predictors — Gradient Boosting. Two GradientBoostingRegressor models (scikit-learn) are trained on the 400×20 matrix:

- Circularity model: $n_{\text{estimators}}=120, \text{max_depth}=4, \text{random_state}=42 \rightarrow$ continuous score clipped to $[1, 5]$.

- Supporting model: $n_estimators=80$, $max_depth=3$, $random_state=42 \rightarrow$ score thresholded at 0.5 for intact/not.

Gradient boosting was chosen because the feature set mixes continuous physics quantities with nonlinear interactions (e.g., high current only becomes catastrophic when strut intersection is large), and tree ensembles handle mixed-scale features without heavy normalization. Depth is kept shallow (3–4) to limit overfitting on a few hundred physically correlated rows derived from only 16 recipes.

(B) Recipe search on SEM labels — Gaussian Process. A `GaussianProcessRegressor` with kernel `Matérn(v=2.5) + WhiteKernel(noise=0.3)`, `normalize_y=True`, is fit on the 16 points $(I, T, D) \rightarrow$ SEM circularity. About 3,000 random candidates are drawn (half concentrated near the gentle island 3.5–5.5 A / 120–150 μ s / 75–80%, half over the full DOE box). Candidates are ranked by $\mu + 0.15 \cdot \sigma$ (mean plus mild exploration). This recovers the Run 4 neighborhood (≈ 3.6 –4.8 A, ≈ 150 μ s, ≈ 79 –80%) with predicted scores ≈ 4.6 –4.9.

(C) Small-n baseline — polynomial Ridge LOOCV. Degree-2 polynomial features of (I, T, D) with `Ridge( $\alpha=0.5$ )` under leave-one-out cross-validation measure how predictable SEM circularity is from recipe alone. Result: LOO MAE ≈ 1.02 on the 1–5 scale — usable as a guide, not as high-precision metrology.

6.5 Training procedure

1. Load `original_16_runs.csv` and `run_visual_labels.csv`.
2. Enumerate landing grid with `grid_positions(step_um=150)`.
3. For each (run, landing): run `analyze_position`, build 20-D $\mathbf{X}$, compute c_{train} and s_{train} .
4. Fit both Gradient Boosting models with `.fit(X, y)`.
5. Persist bundle {circularity, supporting, n_features=20} via `joblib` for reuse at inference.

No random train/test split of landings is used as the primary protocol, because landings from the same run are strongly dependent. Instead, honesty comes from LOOCV on the 16 independent SEM recipes plus physical consistency checks (next subsection).

6.6 Validation and testing strategy

Quantitative: Leave-one-out CV on the 16 SEM circularity labels (Ridge/polynomial) $\rightarrow$ MAE ≈ 1.02 . Exploratory models that include the 1,100 GP synthetics use 5-fold cross-validated MAE on hole-deviation responses to watch overfitting in EDM space only.

Qualitative consistency tests (critical for n=16): (i) for Run 4-like recipes, predicted circularity must peak at low geometry-risk pore-centered landings; (ii) for $I \geq 8$ A, predictions must stay FAIL across the grid; (iii) GP search must place top candidates near 4 A / 150 μ s / 80%; (iv) deviation-minimizing recipes must not outrank SEM-successful ones when the circularity target is used. SEM images remain the final acceptance test.

6.7 Physics heuristic and ML–physics blending at inference

A transparent heuristic score c_H encodes SEM lessons without trees: base ≈ 4.2 –4.5 for gentle recipes ($I \leq 5$, long T , high D); base ≈ 1.8 –2.2 for aggressive recipes; then $c_H \leftarrow \text{clip}(c_H - 2.2 \cdot \text{geometry_risk} + 0.3 \cdot [\text{dist_strut} > 150], 1, 5)$. Supporting heuristic requires gentle EDM and $\text{geometry_risk} \leq 0.55$ (stricter if intersection length is large); $I \geq 8$ always fails.

Final score blends ML and heuristic:

$$c = (1-w) \cdot c_{\text{ML}} + w \cdot c_H, \quad w = \min(0.7, 0.35 + 0.15 \cdot \text{drift}),$$

where $\text{drift} = |\text{tool}\varnothing - 900|/900 + |\text{pore}\varnothing - 235.6|/235.6$. Near the lab geometry, ML dominates; when the user changes tool/pore sizes far from training, the heuristic weight rises so the system does not confidently hallucinate. Supporting integrity similarly requires heuristic agreement when w is large. **Circularity ratio** = $c/5$. PASS if $c \geq 3.5$, supporting intact, and geometry risk ≤ 0.55 .

Stage	Data	Method	Purpose
Recipe search	16 SEM labels	GP Matérn 5/2	Find robust (I,T,D)
Small-n CV	16-run LOOCV	Poly Ridge $\alpha=0.5$	MAE ≈ 1.02 / 5
Densify EDM	GP posterior	1,100 synthetics	Fill input space

Stage	Data	Method	Purpose
Spatial labels	16×25 landings	Risk-modulated SEM	Teach (x,y) effect
Fit models	400×20 features	GBR 120 + GBR 80	Score & support
Inference	Any (I,T,D,x,y)	ML + heuristic blend	Ratio = score/5

Table 2. End-to-end ML stack from labels to inference.

7. Predicting Circularity Ratio for Any Position

For a query (I, T, D, x, y) the runtime path is:

1. **Geometry pass:** `analyze_position(x, y)` computes distances, overlaps, intersection length, and `geometry_risk`.
2. **Feature build:** concatenate 7 EDM features with 13 geometry features $\rightarrow$ 20-D row.
3. **ML forward:** circularity GBR $\rightarrow c_{ML}$; supporting GBR $\rightarrow s_{ML}$ (threshold 0.5).
4. **Heuristic forward:** evaluate c_H and s_H from the gentle/aggressive rules and risk.
5. **Blend:** compute drift-based weight w ; $c = (1-w)c_{ML} + wc_H$; combine supporting flags.
6. **Report:** score c , circularity ratio $c/5$, supporting OK, geometry risk, PASS/FAIL. Repeating steps 1–6 on a fine (x, y) grid yields a spatial circularity field for fixed EDM settings.

8. Final Recommended Parameters

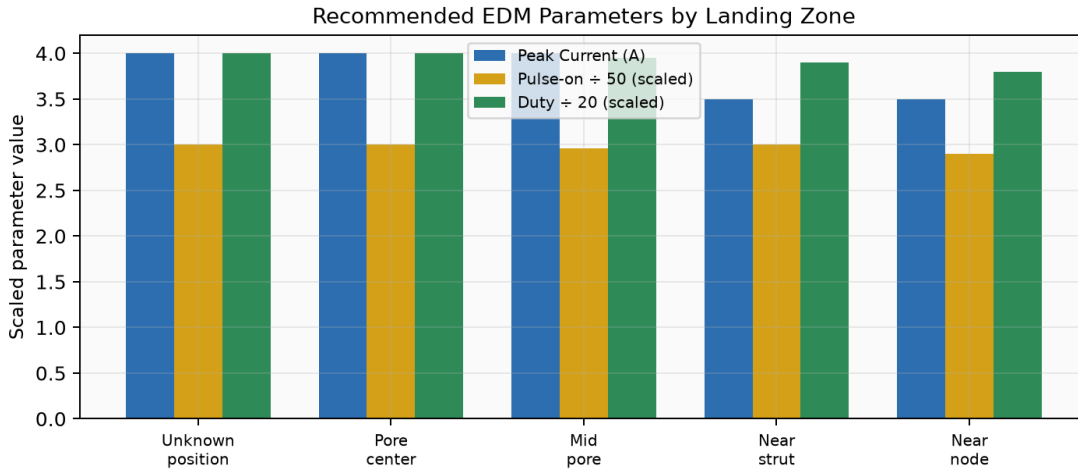


Fig. 12. Recommended EDM settings by landing zone (unknown position uses the robust Run 4 island; near-strut/node landings soften current).

Landing situation	I (A)	T (μ s)	D (%)	Basis
Position unknown (robust)	4	150	80	SEM Run 4 + GP
Pore center	4	150	80	Low geometry risk
Mid pore	4	148	79	Near-center
Near strut	3.5	150	78	Higher geometry risk
Near node / corner	3.5	145	76	Corner overlap
Deviation-only (reject)	6	50	64	Fails SEM ring

Table 2. Final parameter answers from the ML/physics process.

9. Conclusion

Sparse SEM-labeled EDM data can still support position-aware circularity prediction if we couple process physics with lattice geometry, expand 16 labels through GP synthesis and risk-modulated spatial replay, train Gradient Boosting, and regularize with heuristics at inference. Circularity ratio becomes computable for any tool position by teaching the geometry–process interaction that SEM revealed — not by collecting thousands of new trials.

References

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- [5] Project datasets & modules (2026): `original_16_runs.csv`, `run_visual_labels.csv`, `synthetic_1100_points.csv`; `lattice_geometry_engine.py`, `circularity_predictor.py`.